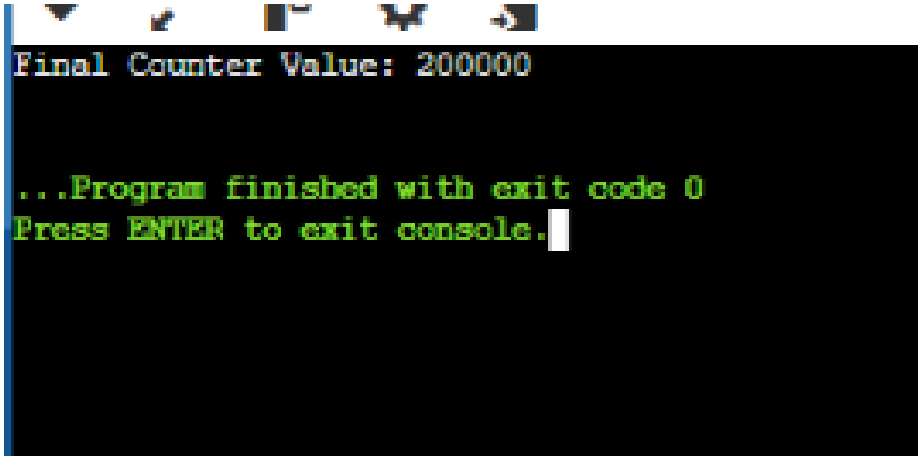
EXPERIMENT 19: Process Synchronization Using Mutex Locks

code:

```
#include <stdio.h>
#include <pthread.h>

int counter = 0;

pthread_mutex_t mutex;

void *threadFunction(void *arg) {
   for (int i = 0; i < 100000; i++) {
      pthread_mutex_lock(&mutex);
      counter++;
      pthread_mutex_unlock(&mutex);
   }
   return NULL;
}

int main() {
   pthread_t thread1, thread2;
   pthread_mutex_init(&mutex, NULL);
   pthread_create(&thread1, NULL, threadFunction, NULL);
```

```
    pthread_create(&thread2, NULL, threadFunction, NULL);

    pthread_join(thread1, NULL);
    pthread_join(thread2, NULL);

    printf("Final Counter Value: %d\n", counter);

    pthread_mutex_destroy(&mutex);

    return 0;
}
```

Output: